Piper Morgan Object Model: Conceptual Decisions Brief

Date: November 27, 2025 (Thanksgiving)
Session: CXO Object Model Exploration
Status: Decisions captured, ready for sketching

Purpose

This brief captures the conceptual decisions made during the object model exploration session. These decisions form the foundation for sketching exercises and subsequent design work. They are "decided for now"—firm enough to build on, open to revision through the sketching process.

Core Architectural Decisions

1. Lenses on Substrates (not discrete object types)

Decision: Piper's conceptual model is organized around a small set of substrates viewed through analytical lenses, rather than a rigid taxonomy of object types.

Rationale:

- More flexible, less rigid categorization
- Avoids false dichotomies ("is this a Task or a Risk?")
- Simpler core model with richer views
- Prevents premature ontological commitment

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2. Four Substrates

Decision: The fundamental "stuff" Piper perceives and reasons about:

Substrate	What It Is	Examples
Entities	Discrete things with identity	People, Work Items, Documents, Artifacts
Spaces	Containers/contexts where work happens	Projects, Channels, Repos, Teams
Moments	Points or spans in time	Events, Deadlines, Sessions, Milestones
Situations	Configurations of the above at a given time	"Current state of the launch," "What's blocked right now"

Rationale: Minimal set that can compose into anything richer. Entities are the nouns, Spaces are the containers, Moments are the temporal anchors, Situations are the snapshots.

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3. Eight Spatial Dimensions = The Lenses

Decision: The 8-dimensional spatial intelligence framework already built serves as the analytical lenses applied to any substrate.

Dimension	What It Illuminates
TEMPORAL	When? Sequence, deadline, history, pattern over time
HIERARCHY	What contains this? What does it contain?
PRIORITY	How urgent/important? What deserves attention?
COLLABORATIVE	Who's involved? Who's blocked? Who cares?
CAUSAL	What caused this? What does it affect?
CONTEXTUAL	What situation makes this relevant?
RELATIONAL	How does this connect to other things?
IDENTITY	What is this? How does Piper recognize it?

Rationale: One model, not two. The intents align; doing otherwise spawns unclear parallelistic complexity. Also helps resist "flattening" the spatial intelligence work.

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4. Three-Way Ownership Split

Decision: Objects are categorized by Piper's relationship to them:

Category	Piper's Role	Metaphor	Examples
Native	Creates, owns, maintains	Piper's Mind	Sessions, Memories, Concerns, Trust States, Orientation, Journals
Federated	Observes, queries, acts upon (doesn't own)	Piper's Senses	GitHub Issues, Slack Messages, Calendar Events, Notion Docs
Synthetic	Constructs from federation + native reasoning	Piper's Understanding	Projects (assembled), Risks (inferred), Patterns (learned)

Rationale: Clear ownership boundaries prevent confusion about source-of-truth. Native = mind, Federated = senses, Synthetic = understanding.

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5. Six Common Metadata Dimensions

Decision: A universal metadata schema applies across ALL objects regardless of type or category:

Metadata	What It Captures
Provenance	Where did this come from? Source, confidence, freshness
Relevance	Why does this matter now? Connection to current context
Attention State	Has this been seen? Does it need attention?
Confidence	How certain is Piper about this?
Relations	What is this connected to? Graph position
Journal	History of Piper's interaction with this object

Rationale: These let Piper answer: Where from? Why now? Seen it? Sure about it? What's it connected to? What's the story? Journal as metadata means every object carries its own history (audit trail, recovery, temporal dimension).

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6. Full Lifecycle with Composting

Decision: Objects move through a complete lifecycle that includes endings and succession:

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Emergent → Derived → Inferred → Proposed → Ratified → Deprecated → Archived → Composted
  ↑                                          |
  └──────────────────────────────────────── feeds new
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Stage	What It Means
Emergent	Piper notices something, hasn't interpreted it yet
Derived	Piper connects it to other things
Inferred	Piper forms a hypothesis about what it is
Proposed	Piper surfaces it to user for consideration
Ratified	User confirms, names, or acts on it
Deprecated	Superseded, no longer current, but retained
Archived	Ended, closed, completed, retired
Composted	Decomposed into learnings that feed new things

Rationale:

- The shadow side of PM work is ending things—model should honor that
- Nothing truly disappears; it transforms
- Composting is where Memory and Pattern objects get fed
- Dreaming/digestion = partly composting
- Cycle closes: death feeds new life

7. Two Journaling Layers

Decision: Piper maintains two distinct journal layers with different visibility:

Layer	Visibility	What Goes Here	When It Surfaces
Session Journal	User can query anytime	What happened, what was produced, the audit trail	On request
Insight Journal	Surfaces when ready	Outputs of dreaming/composting—patterns noticed, premonitions, connections made	At appropriate trust level, or on request

Not formalized: Private diary (internal working state) remains internal state, not a journaled layer.

Rationale:

- Not everything in cognition should be visible immediately
- Insight Journal is where "composted" material surfaces as new insight
- Avoids optional complexity of three layers
- Private processing happens but doesn't need to be journaled

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8. User Model as Native Object (Principal + Others)

Decision: Piper maintains explicit models of humans, with depth varying by proximity:

Level	Who	Model Depth	Core Dimensions
Principal	Account holder Piper works for	Deep	Wants, fears, preferences, patterns, trust grants, goals, stress indicators
Team	Humans principal works with directly	Medium	Role, relationship to principal, communication style, what they care about
Stakeholders	Humans referenced in work context	Light	Wants, fears (stakeholder mapping), relationship to work
Mentioned	Names that appear but aren't active	Minimal	Identity and context where mentioned

Core heuristic for modeling humans: "What do they want?" (what's a win for them) and "What are they afraid of?" (what has burned them, status risks)

On creepiness: The antidote is transparency, correctability, forthrightness, and clear boundaries. Kind, not just nice.

Parked: Non-human sapients (other AI agents) may need modeling eventually but not MVP-critical.

What This Enables

With these decisions, Piper has:

- **A way to perceive** (substrates + lenses)
- **A way to categorize relationships** (native/federated/synthetic)
- **A way to track anything** (common metadata)
- **A way to understand change** (lifecycle with composting)
- **A way to remember and reflect** (two journal layers)
- **A way to understand humans** (user model with wants/fears)

Open for Sketching Exploration

The following remain open for discovery through visual exploration:

- How do substrates, lenses, and ownership categories actually compose visually?
- What does the lifecycle look like as a diagram? (Circular? Spiral? Branching?)
- How do the six metadata dimensions attach to objects?
- What does the user model hierarchy look like in relation to work objects?
- How do the two journal layers relate to the lifecycle (especially composting)?

Metaphor Summary

Concept	Metaphor
Native/Federated/Synthetic	Mind / Senses / Understanding
Lifecycle ending	Composting (death feeds new life)
Piper's role	Junior PM / Assistant with a principal
User modeling	Wants and fears (stakeholder empathy)
Journaling	Session (audit) + Insight (dreams surfaced)
Colleague test	"Would a good PM colleague know this?"

Document Status: Complete snapshot of decisions
Next Step: Sketching assignment for Object Model visualization

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Captured: November 27, 2025, ~3:38 PM Pacific